

CAPSTONE PROJECT • SECUREGUARD FINANCIAL SOLUTIONS

Financial Fraud Detection

Anomaly Detection in Credit Card Transactions

A multi-tool analysis using Excel, SQL, Python and Tableau to flag suspicious transactions in real time

389,002

TRANSACTIONS ANALYZED

2,252

FRAUD CASES FOUND

0.58%

FRAUD RATE

Why Fraud Detection Matters

SecureGuard Financial Solutions builds real-time fraud detection systems for the financial industry. As online payments grow, so does the risk of unauthorized transactions, identity theft and unusual spending patterns — this project analyzes transaction data to surface those anomalies.

PROBLEM STATEMENT

Design a fraud detection system to identify fraudulent credit card transactions in real time — detecting unauthorized transactions, unusual spending patterns and fraudulent card usage to prevent financial losses for cardholders and institutions alike.

Approach: analyze the cc_data transaction dataset across four tools to explore, quantify and visualize fraud patterns.

THE STAKES



Financial loss

Fraudulent charges drain both cardholder and bank funds.



Trust erosion

Repeated fraud damages customer confidence in a bank.



Scale of data

Millions of transactions must be screened in real time.

A Four-Tool Analytical Workflow

Each tool tackled a different layer of the fraud detection problem.

1

Excel

Statistical summaries, histograms, pivot reports on fraud by gender & category, correlation checks.

2

SQL

Schema design, table joins, aggregation queries to quantify fraud volume, merchants & spending patterns.

3

Python

Full EDA — distributions, missing values, outliers, correlation with the fraud target variable.

4

Tableau

Interactive dashboards: box plots, geographic fraud maps, and time-series transaction trends.

The Data at a Glance

389,002

TRANSACTIONS

22

VARIABLES

14

MERCHANT CATEGORIES

693

UNIQUE MERCHANTS

51

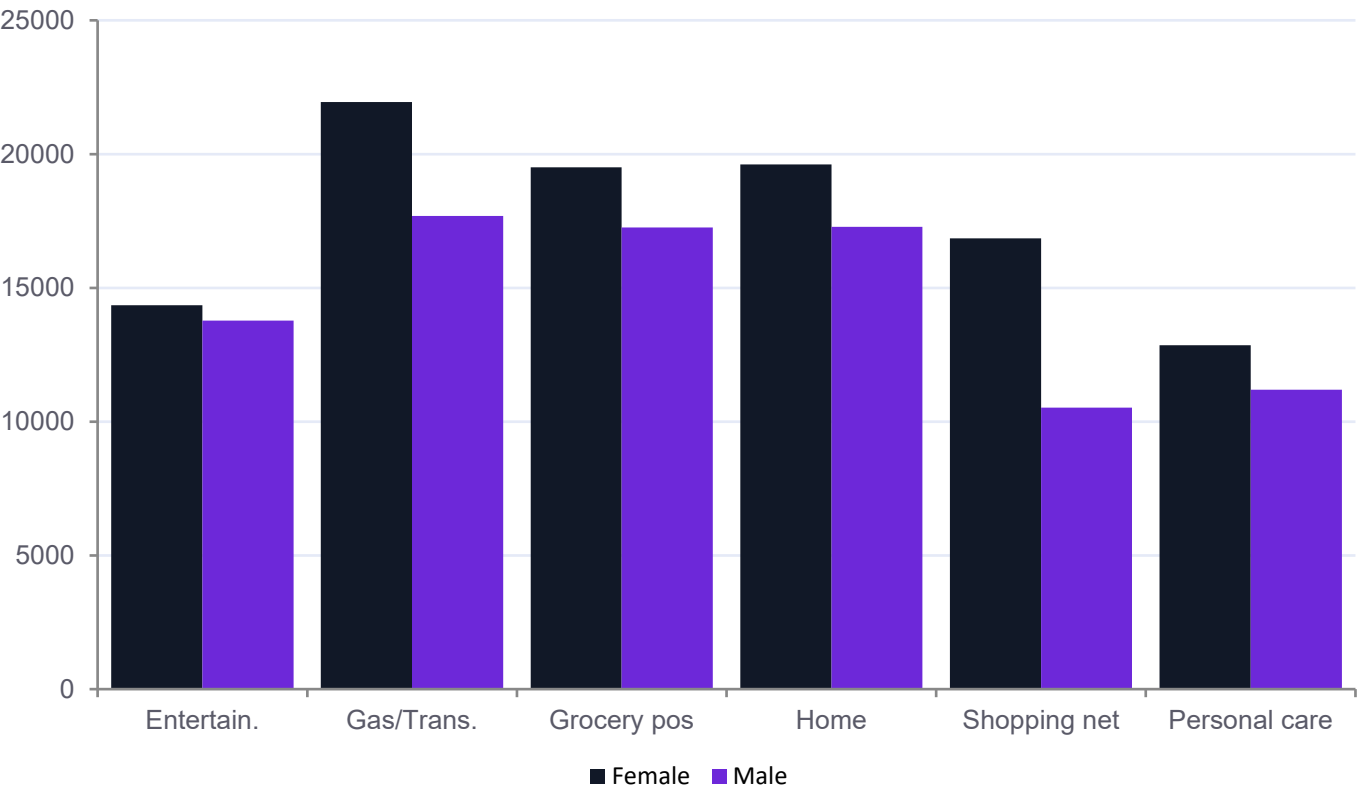
STATES COVERED

0

MISSING VALUES

Transactions by Gender & Category

A pivot report breaks down all 389,002 transactions by gender across 14 spending categories.



EXCEL TASK

Report: Frauds by Gender & Category

Built with a PivotTable on gender × category, counting all transaction records.

TOTALS

Female: 213,231 transactions

Male: 175,771 transactions

Grocery (pos), gas/transport and home dominate spending volume for both genders.

Geography & Correlation Checks

TOP 3 STATES BY TRANSACTION VOLUME



These three states alone account for 77,427 transactions in the sample.

AVG. TRANSACTION AMOUNT BY JOB ROLE

492 distinct job roles were compared — most cluster near the \$70–90 average, with a handful of high-value outlier roles (e.g. air traffic controller, \$503.75).

CORRELATION CHECK

Transaction Amount vs. City Population

$$r = 0.0073$$

Essentially no linear relationship — how much a customer spends is independent of how big their city is. City population alone is not a useful predictor of transaction size.

Implication: fraud models should lean on transaction-level features (amount, category, timing) rather than demographic proxies like city size.

Querying the Transaction Database

A finance schema joins `cc_data` and `location_data` to quantify fraud and spending behavior.

FRAUD VOLUME & SHARE

```
SELECT SUM(is_fraud) AS fraud_transactions,  
       ROUND((SUM(is_fraud) /  
              COUNT(*)) * 100, 2)  
       AS fraud_percentage  
FROM cc_data;
```

2,252 fraudulent transactions • 0.58% of all activity

TOP MERCHANTS & CATEGORY SPEND

GROUP BY merchant / category, ranked by transaction count and average amt — surfaces the busiest merchants and which categories run the highest average ticket size.

PEAK SPENDING DAY

```
SELECT DAYNAME(trans_date_trans_time)  
       AS day_of_week,  
       AVG(amt) AS avg_transaction_amount  
FROM cc_data  
GROUP BY day_of_week  
ORDER BY avg_transaction_amount DESC  
LIMIT 1;
```

Identifies the single day of the week with the highest average ticket size — useful for staffing fraud review teams around peak-risk periods.

Distributions, Outliers & Data Quality

0

Zero Missing Values

Every one of the 22 fields is fully populated across all 389,002 rows.

40.3

Right-Skewed Amounts

Skewness of the amt field — a long tail of large, unusual transactions.

20,377

Amount Outliers

Transactions flagged by the IQR method as unusually large or small.

72,813

City-Pop Outliers

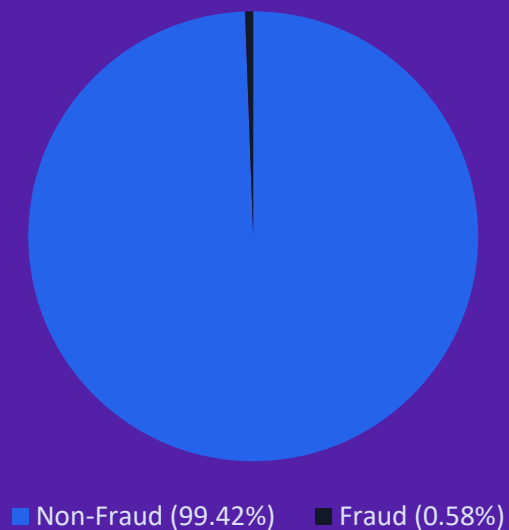
Records from cities whose population sits far outside the typical range.

WHY NOT JUST DELETE THE OUTLIERS?

Unusually large transactions and population values were kept rather than removed — in fraud detection, the 'outliers' are often exactly the signal the model needs to learn from.

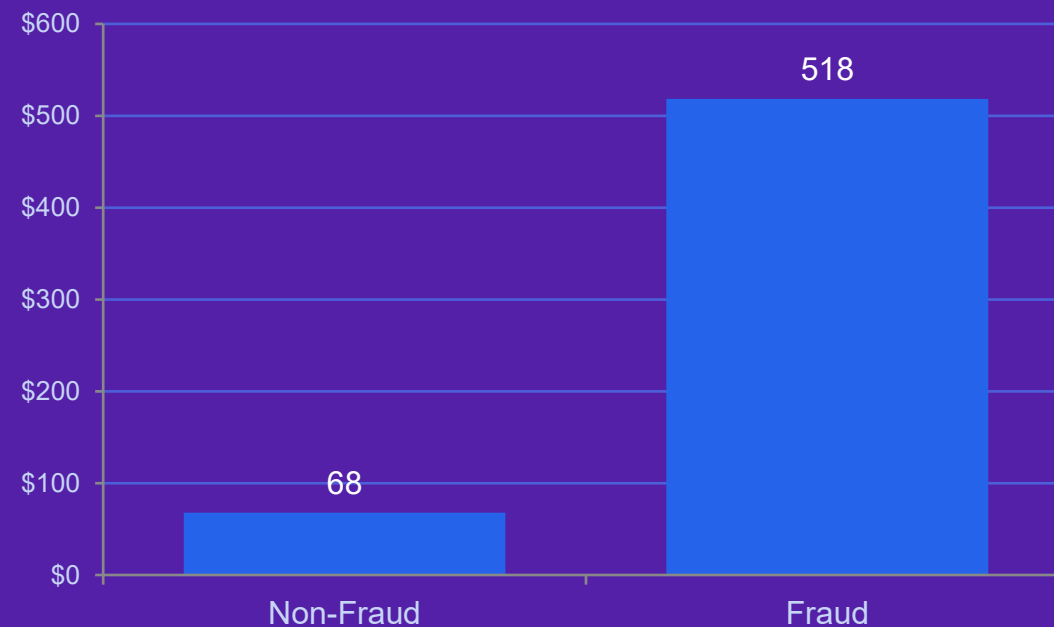
What Makes a Transaction Fraudulent?

CLASS IMBALANCE



Only 2,252 of 389,002 transactions are fraudulent — any model must handle this severe imbalance.

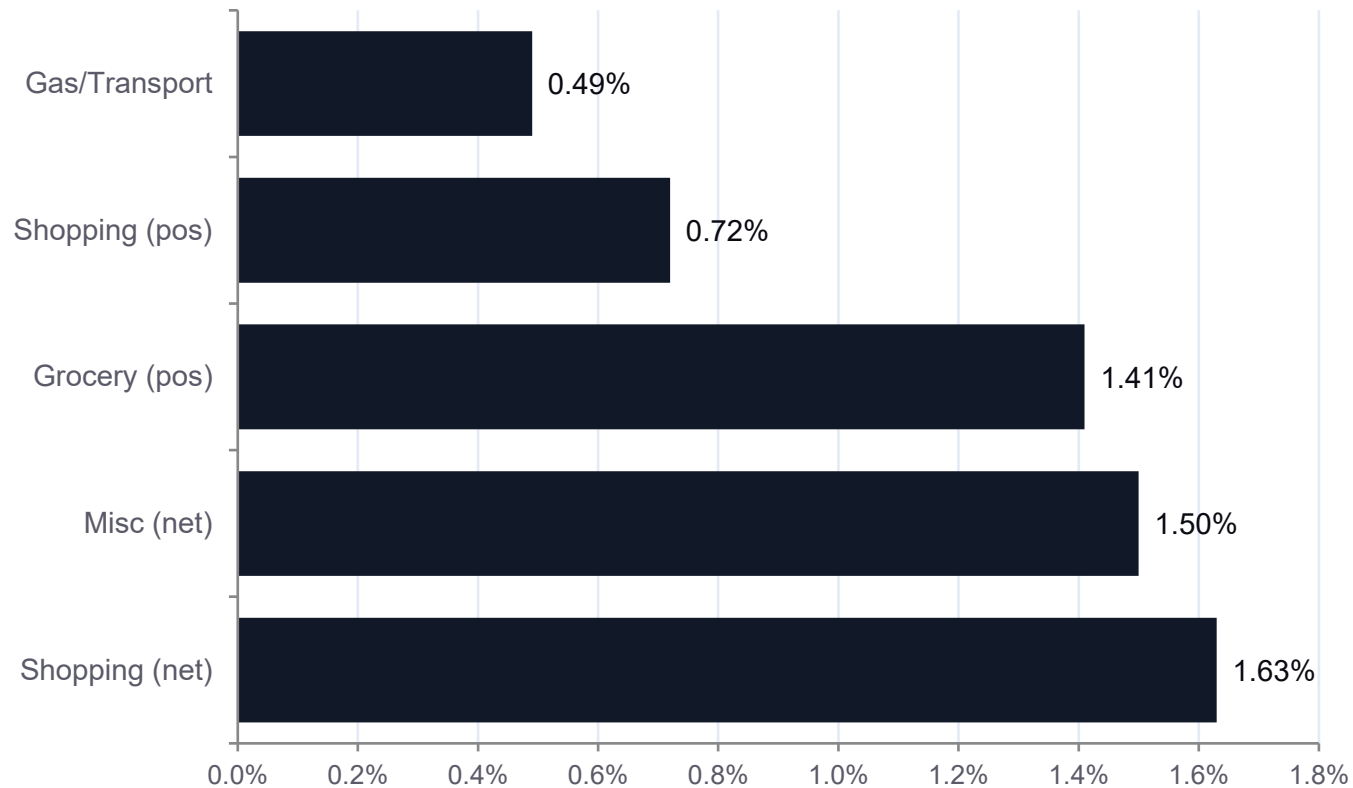
AVERAGE TRANSACTION AMOUNT



Fraudulent transactions average \$518 — about 7.6× higher than genuine ones (\$68), the single strongest numeric signal ($r = 0.21$ with `is_fraud`).

Fraud Rate by Category

Digital-first categories carry the highest fraud rates — physical, routine purchases the lowest.



SQL TASK

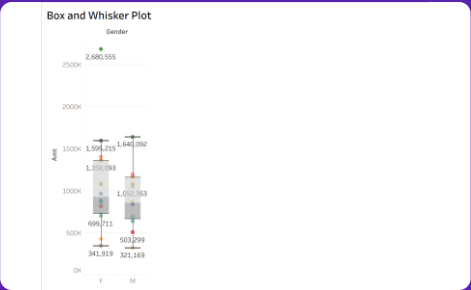
Fraudulent transactions per category

```
grocery_pos: 518
shopping_net: 478
misc_net: 287
shopping_pos: 253
gas_transport: 193
```

Raw fraud counts (left) vs. fraud rate as a share of category volume (chart) tell different stories — grocery_pos has the most fraud cases, but shopping_net has the highest rate.

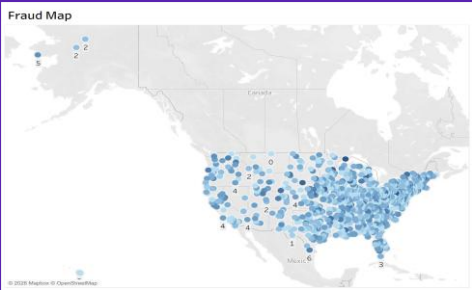
Visualizing Fraud Interactively

Four linked worksheets built from cc_data.csv, designed to be explored interactively in Tableau.



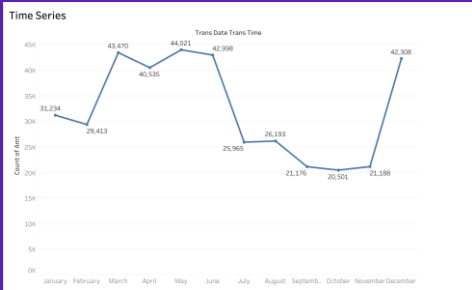
Box & Whisker Plot

Amount spent by gender across every category, showing spread and outliers side by side.



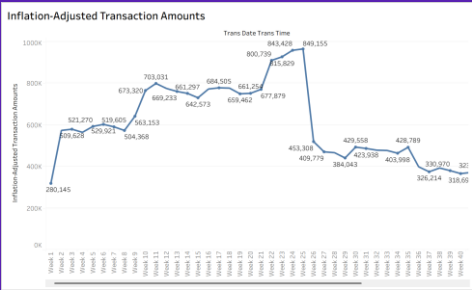
Fraud Map

Latitude/longitude plot colored by Is Fraud, pinpointing where flagged transactions cluster.



Time Series

Monthly transaction count trend line, revealing seasonality in overall activity.

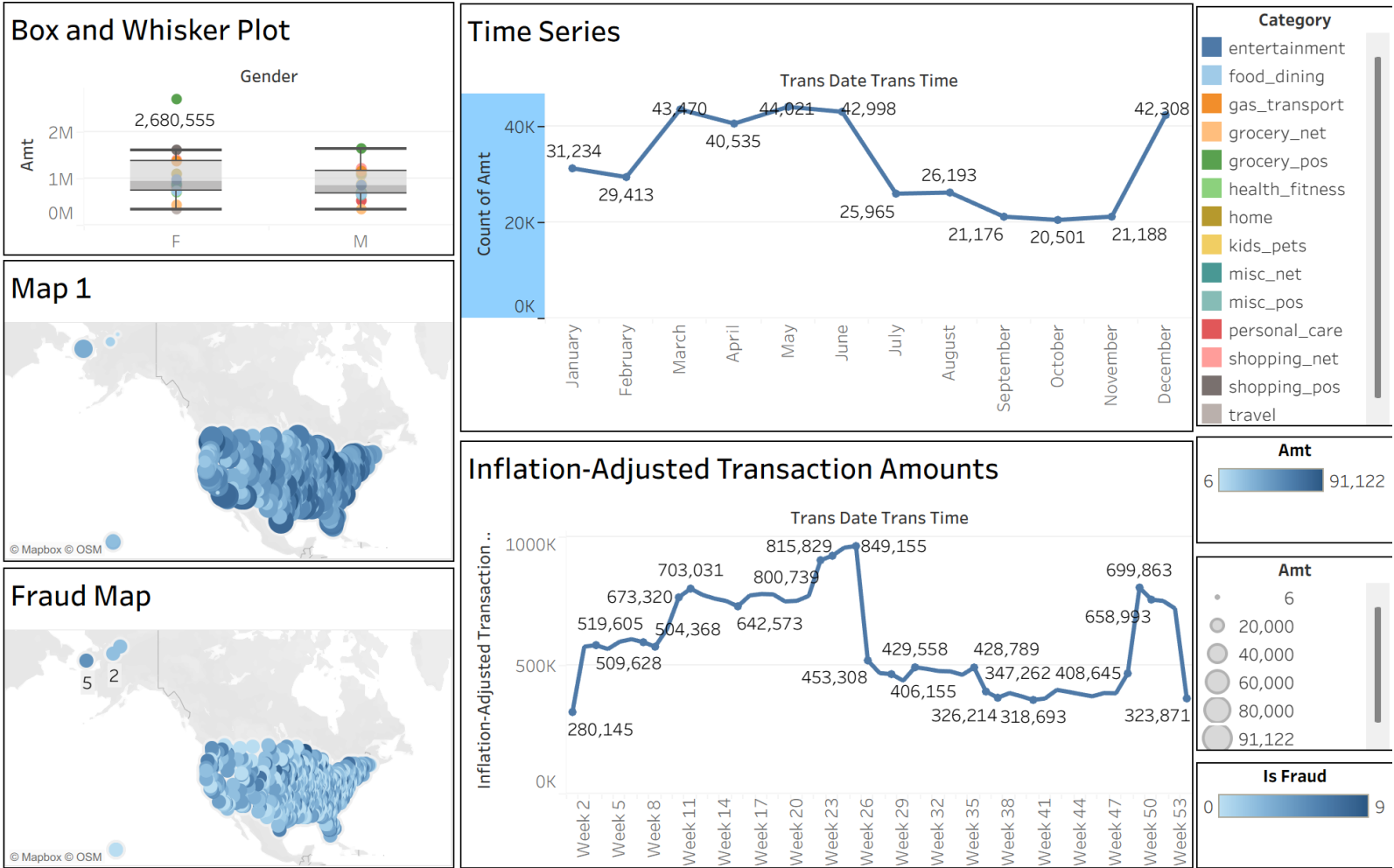


Inflation-Adjusted Amounts

Weekly inflation-adjusted transaction totals, tracking real purchasing-power shifts over the year.

The Complete Dashboard

All four views combined into a single interactive Tableau dashboard.



What the Analysis Reveals

1

Fraud is rare but severe: just 0.58% of transactions are fraudulent, yet they average 7.6× the transaction size of genuine purchases.

2

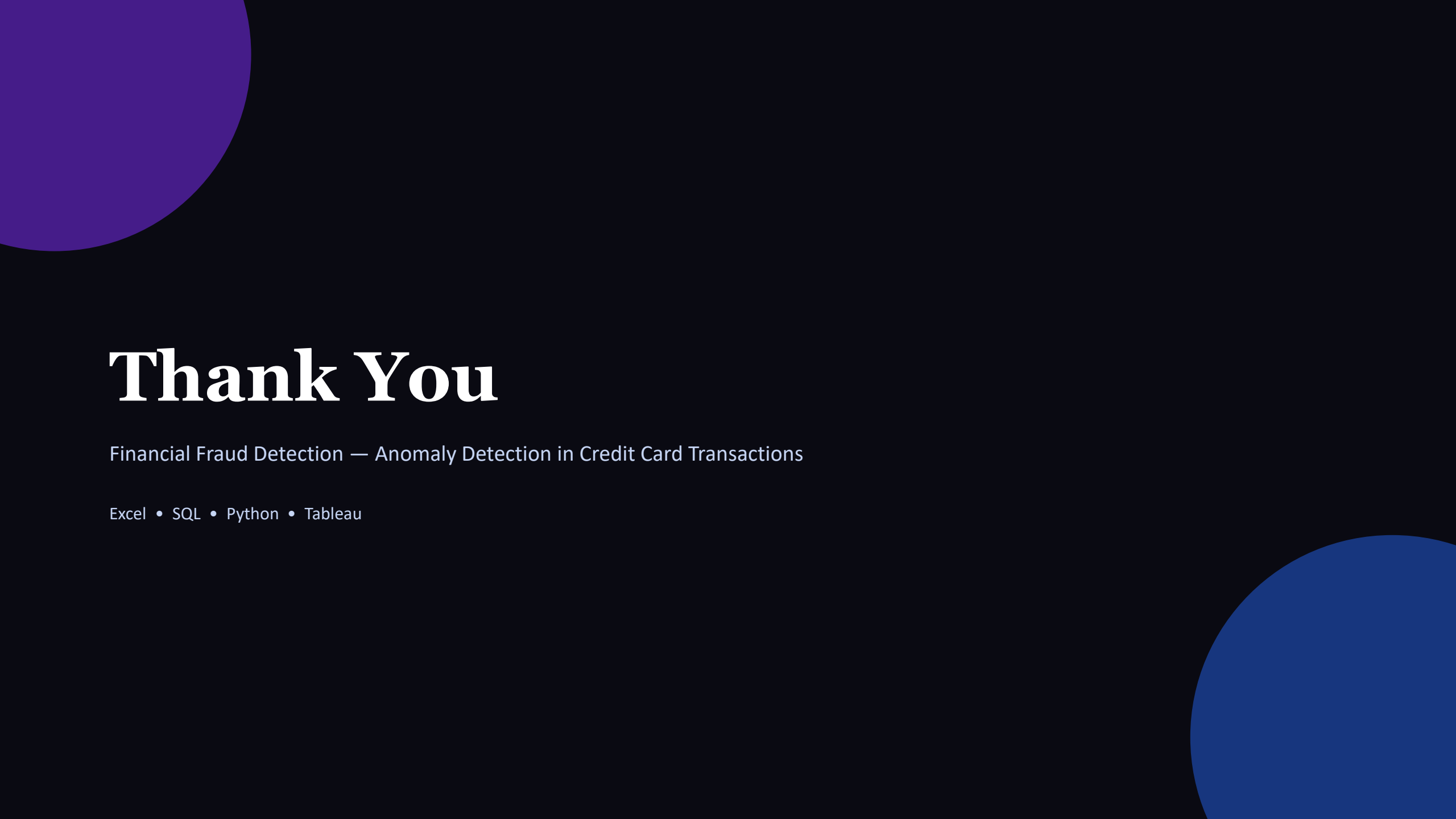
Transaction amount is the strongest numeric predictor of fraud ($r = 0.21$) — far ahead of location, age or city population (all near-zero correlation).

3

Digital-first categories — online shopping and misc. net purchases — carry the highest fraud rates, while routine in-person spending is comparatively safe.

4

The dataset is clean (zero missing values) but heavily skewed, with tens of thousands of legitimate outliers that a fraud model should learn from, not discard.



Thank You

Financial Fraud Detection — Anomaly Detection in Credit Card Transactions

Excel • SQL • Python • Tableau